

MCV4UY - Unit 2 Test - Limits and Continuity

Name _____

Date _____

Total
/40

KNOWLEDGE

1. Evaluate the following limits. Use the method of your choosing.

a. $\lim_{x \rightarrow \infty} \frac{2x+1}{\sqrt{x^2+3}}$

b. $\lim_{x \rightarrow 3} \frac{\sqrt{x+6}-x}{x^3-3x^2}$

(2)

(3)

c. $\lim_{x \rightarrow 0} \left\{ \frac{1}{x} \left[\frac{1}{(2+x)^2} - \frac{1}{4} \right] \right\}$

d. $\lim_{x \rightarrow 0} \frac{\tan 5x}{\sin 3x}$

(2)

(3)

2. For $f(x) = \frac{x^2+3x-10}{2x^2-7x+6}$

a. Determine the vertical asymptote(s) and describe the function's behavior of x as x approaches from the left and right of each asymptote.

(2)

b. Determine the horizontal asymptote(s).

(1)

c. State the domain of the function.

(1)

APPLICATION

3. Given the function $f(x) = \begin{cases} x^2 \sin\left(\frac{3\pi}{4}x\right) & x < 2 \\ ax^2 + bx - 18 & 2 \leq x < 5 \\ x^2 - b & x \geq 5 \end{cases}$, determine the values of a and b such that the function is continuous throughout.

(4)

4. The volume V of the cylinder is represented by the function, $V = V(r) = 5\pi r^2$, where r represents radius in cm.
- a. Determine an equation that represents the instantaneous rate of change of the volume with respect to the radius.

(2)

- b. At what radius will the cylinder have an Instantaneous rate of change of 55π .

(1)

5. Use the Squeeze theorem, to determine the $\lim_{x \rightarrow 0} \left[\sqrt{x^3 + 3x^2} \sin\left(\frac{1}{x}\right) \right]$

(2)

COMMUNICATION

6. Explain using the **Intermediate Value Theorem**, how to determine whether $2x^3 + 3x^2 - 23x - 42 = 0$ has a zero in the interval $[3, 4]$.

(2)

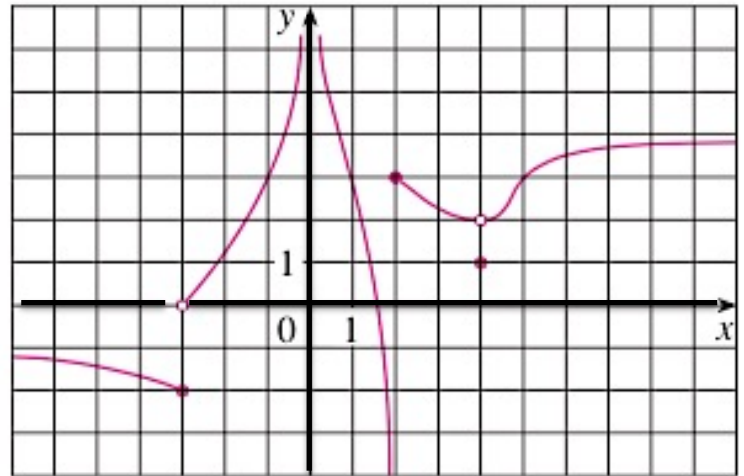
7. a. Observe the function $f(x)$ shown on the graph and determine the followings limits:

i. $\lim_{x \rightarrow -3^-} f(x) =$

ii. $\lim_{x \rightarrow 0} f(x) =$

iii. $\lim_{x \rightarrow -2^-} f(x) =$

iv. $\lim_{x \rightarrow 4^+} f(x) =$



(2)

b. Determine the value(s) of x where the function is discontinuous. Explain why the function is discontinuous at these values.

(2)

THINKING

8. Find the constants **a** and **b** such that $\lim_{x \rightarrow 0} \frac{\sqrt{ax+b}-2}{x} = 1$.

(3)

9. Determine the $\lim_{x \rightarrow 0} \frac{|2x-1|-|2x+1|}{x}$.

(3)

10. Find the equations of the tangents of $f(x) = x^3 + x^2 + x - 1$ that are normal to the line $x + 2y + 5 = 0$.

(4)